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GRE Probability & Counting Theory

1 Counting

In this section, we will discuss counting the number of ways to select or arrange a large number of items.

In counting, as we will in probability

1. "and" means multiply
2. "or" means add

1.1 Worked Examples

- If, at a fancy dinner, you can choose one of three meat entrees or one of two vegetarian entrees, how many choices do you have?

There are, in total, five entrees from which to choose. $5 = 3 + 2$

- Instead, if you can choose one of three entrees and one of two desserts, how many entree + dessert combinations are there?

We could have any of three entrees with dessert #1, and any with dessert #2, so that's six possibilities. $6 = 2 \times 3$

Another important idea is listing the items you need to count.

2 The Fundamental Counting Principle (FCP)

The FCP is a general way to approach tasks that can be broken into stages.

Suppose we can divide a given task into stages.

Suppose the first stage can be done in n_1 , the second in n_2 ways, and so forth.

The total number of ways to do the task will simply be the product of these numbers.

$$N = n_1 * n_2 * \cdots$$

2.1 Worked Examples

- For a formal dinner, guests have the choice of one of 4 salads, one of 5 appetizers, one of 12 entrees, and one of 4 desserts. How many different meals are possible?

$$N = 4 * 5 * 12 * 4 = 960$$

- Suppose we have six different books that we will place on a shelf. In how many different orders can we place these books?
 - first book =6
 - second =5
 - third =4
 - fourth=3
 - fifth =2
 - sixth=1

$$n = 6 * 5 * 4 * 3 * 2 * 1 = 720$$

As a general rule, if we have to arrange n different items in order, the total number of orders is the product of n and every positive integer less than n .

2.2 Practice problem:

A small division of a company, with 25 employees, will choose a three-person steering committee consisting of a facilitator, a union rep, and a secretary. How many different steering committees could be chosen?

$$F = 25, UR = 24, S = 23$$

$$N = 25 * 24 * 23 = 13,800$$

Many counting problems contain a restriction of some kind.

When a restriction of this sort appears in a counting problem that can be broken into stages, always start with the most restrictive stage first.

2.3 Worked Examples:

- The children AL, Betty, Chuck, Dahlia, Ed, Fran and George will sit in seven adjacent chairs. Dahlia must be in the middle chair, and George must be next to Dahlia. In how many orders can they be arranged?

$$D=1; G=2; A=5; B=4, C=3; E=2; F=1$$

$$N = 1 * 2 * 5 * 4 * 3 * 2 * 1 = 240$$

- In a certain summer school program, there are five periods in the day. Each student takes English, Math, Science, and Science Lab. In how many orders can a student schedule be arranged, given that Science Lab must immediately follow Science class?

$$S=4; SL=1; E=3; M=2; H=1/$$

$$N = 4 * 1 * 3 * 2 * 1 = 24$$

3 Factorial Notation

The pattern of n times all the positive integers less than n is called " n factorial".

$$5! = 5 * 4 * 3 * 2 * 1 = 120$$

Factorials get very big very quickly.

- $10! > 1$ million
- $13! > 1$ billion

We have learned that if we have n different items, we can put these items in $n!$ different orders.

Think: every arrangement of the total $n!$ must either obey the restriction or not.

- $n!$ = total # of arrangements
- R = # of arrangements that obey the restriction
- Q = # of arrangements that do not obey

$$n! = R + Q$$

Therefore $R = n! - Q$

3.1 Worked example

- Six children will sit in a row of six chairs, but Jackie and Marilyn cannot be seated next to each other. How many arrangements are possible?

$$\text{Total} = n! = 720$$

Let's count the arrangements that don't meet this restriction, that is, all the arrangements with J & M next to each other.

There are 10 arrangements where they sit next to each other. The four empty seats could be filled by $4! = 24$

$$\text{So, not allowed} = 10 * 24 = 240$$

$$\text{allowed} = 720 - 240 = 480$$

- Harriet has six novels she wants to read, one of which is Emma by Jane Austen. She plans to create a reading list of four of these novels for an upcoming trip, and different orders count as different lists. How many reading lists are possible if Emma must be on the list?

$$\text{Total number of lists} = 6 * 5 * 4 * 3 = 360$$

Without Emma, we can make an ordered list of the five remaining books.

$$\text{Lists without Emma} = 5 * 4 * 3 * 2 = 120$$

$$\text{List with Emma} = 360 - 120 = 240$$

Sometimes, we have to arrange sets in which some of the items are identical, and this requires a special approach to calculation.

3.2 Worked Example:

- A librarian has seven books to arrange: four different novels, and three identical copies of the same dictionary. In how many different orders could these seven books be put on the shelf?

There are $3!=6$ ways to rearrange the 3 dictionaries

$$N = \frac{7!}{3!} = \frac{7 * 6 * 5 * 4 * 3 * 2 * 1}{3 * 2 * 1} = 840$$

Notice that in a set of n items, there are b identical items, and the total number of arrangements is

$$N = \frac{n!}{b!}$$

- How many different arrangements can we make of the eleven letters in the name of the US state Mississippi?

$$N = \frac{11!}{4!4!2!}$$

- A librarian has 5 identical copies of book A, two identical copies of book B, and a single copy of book C. In how many distinct orders can the 8 books be arranged on a shelf?

$$N = \frac{8!}{5!2!} = \frac{8 * 7 * 6}{2} = 168$$

Now we will analyze some of these situations from the perspective of the FCP.

3.3 Worked Examples

- Suppose there is a room of 20 people, and each one will shake hands with each other person. How many handshakes occur?

One might think the number would be $20*19$, but that would be too big - it would count every handshake twice. We have to divide that by 2.

- In geometry, a diagonal of a polygon is a segment from any vertex to any other non-adjacent vertex. For example, a diagonal from K could go to any vertex from A to I, but not to J or L. How many diagonal does a regular dodecagon have?

Choose any starting vertex (12). Then, any different, non-adjacent vertex (9).

$$N = (12 * 9)/2 = 54$$

- From a set of ten different items, Lisa is going to select three as a gift for someone. How many different sets of three can she pick ?

$$N = \frac{10 * 9 * 8}{6} = 120$$

4 Combinations

4.1 Theory

- Suppose we have 6 employees of equal rank in a work group, and we are going to choose randomly 3 of them for a task.

Let's say the 3 chosen are {B, E,F}. All that matters is who has been selected. It doesn't matter at all which one was picked first or second

In a combination, only the final selected group matters; the order in which they were selected for that group doesn't matter at all.

If we have a group of n different individuals, and we randomly select r of these individuals, then the number of combinations is denoted as

$$nC_r$$

This is often read as "n choose r".

- $8C_3$ would be the number of different 3-person combinations we could select from a pool of eight.
- $nC_1 = n$
- In general, if we select a group of r from a pool of n , we automatically create a group of $(n-r)$ folks who were left behind. Therefore

$$nC_r = nC_{(n-r)}$$

4.2 Worked Examples

- How can we calculate $10C_4$?

$$N = \frac{10 * 9 * 8 * 7}{4!} = 210 = 10C_6$$

- An amusement park has 12 different major rides. A coupon gives its holder access to any 3 of these rides for free. How many sets of three rides are possible?

$$N = \frac{12 * 11 * 10}{3!} = 220$$

Sometimes, on more advanced problems, you may need to know the abstract combinations formula:

$$nC_r = \frac{n!}{r!(n-r)!}$$

We use combinations when order doesn't matter.

If order matters in a problem, then most likely we will break the selection problem into stages, count the possibilities in each stage, and use the FCP.

If order doesn't matter, it's a combination, and we will use the combinations approach. (Start with FCP, divide off the repetitions.)

4.3 Worked Examples

- Suppose we have a problem in which we have a pool of 20 committee members, and for the leadership team, we are going to pick three officers: a chairperson, a treasurer, and a secretary. Does order matter?

If we can swap the members of this selected group and this meaningfully changes the outcomes, then order matters.

$$N = 20 * 19 * 18$$

- Suppose a chef is making a soup and has 20 vegetables from which to choose. He is instructed to choose any three vegetables, cut them into small pieces, and stir them into the large pot of soup. How many different soups could he make?

$$N = \frac{20 * 19 * 18}{3!} = 1140$$

- A classical concert will consist of two overtures in the first half and one symphony in the second half. A symphony and two overtures determine a program. The conductor can choose the program from 10 overtures and from 6 symphonies. How many problems are possible?

For the overtures, there are $10C2$ possibilities, and for the symphonies, there are $6C1$ possibilities.

$$N = 6 * \frac{10 * 9}{2!} = 6 * 45 = 270$$

- A librarian has a set of ten books, including four different books about Abraham Lincoln. The librarian wants to put the ten books on a shelf with the four Lincoln books next to each other in the shelf. How many different arrangements of the ten books are possible?

Since the four Lincoln books must be together, think of them as one big book. In how many ways can we put these seven books in order on a shelf? $7!$

That method of counting gives us a position for the Lincoln books among the other six books, but we still can put the four Lincoln books in any order. $4!$

$$N = 7! * 4!$$

5 Permutations and Combinations

One useful permutation formula:

For the permutation of n different items, # of permutations $= n!$

5.1 Worked Example

From a group of 10 committee members, three will be randomly selected: one as chairman, one as treasurer, and one as secretary. These three will form the "board" of the committee. How many different possible "boards" can be formed?

From FCP:

$$N = 10 * 9 * 8 = 720$$

That problem was a "permutation" problem because "order mattered".